Gursewak Singh

Karnal, Haryana, India
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Date of Birth: 03/10/2002

Education

MSc. Physics - I 2024 – 2026
Mumbai University/SVKM's Mithibai College, Mumbai
Marks Sem I: 513/700
Marks Sem II: 434/600

BSc Physics, Chemistry, Mathematics 2020 – 2024
Kurukshetra University/Pt. CLS. Government College, Karnal
Overall Marks : 2022/2900

Senior Secondary Examination 2019 - 2020
Government Model School, Taraori, Karnal
Overall Marks: 371/500

Secondary Certificate Examination 2017 - 2018
Government Model School, Taraori, Karnal
Overall Marks: 383/500

Skills

Programming: Python (NumPy, Matplotlib, SciPy)

Tools: Latex, Excel, Mathematica

Languages Punjabi (Native), English, Hindi.

Other: Basic graphic design and 2D animation (frame-by-frame)

Training, Workshops, and Certifications

Gravitational Wave Open Data Workshop May 2025
Attended the Gravitational Wave Open Data Workshop focused on data analysis techniques, and interpretation of gravitational wave data using Python based tools.

Winter School | National Astronomical Observatory of Japan Feb 2025
Covered topics like stellar evolution, galaxies, and time-domain astronomy.

Winter School | IISER Berhampur Bookworms Club Dec 2024
Covered entropy, astrobiology, and basic life-related physics.

Data-Driven Astronomy (Coursera)
Used Python to explore and analyze astronomical data.

Research Experience

Research Intern, Physical Research Laboratory (PRL), Ahmedabad May – June 2025
Project: Destruction Mechanism of Presolar Grains

I worked on the destruction of presolar grains in extreme environments such as supernova explosions. I used Python to build simulations and understand how different factors like temperature, speed, and grain size affect the survival of dust grains. I also tested my models using data from research papers. This project helped me learn more about space science, coding, and how dust behaves in the universe.

Research Interests

My primary interests lie in Cosmology and Astrophysics, with a focus on analytical methods, numerical methods, and the study of cosmic phenomena.